

Digital Twin Dashboard with Real-Time Robot Telemetry

Team Members

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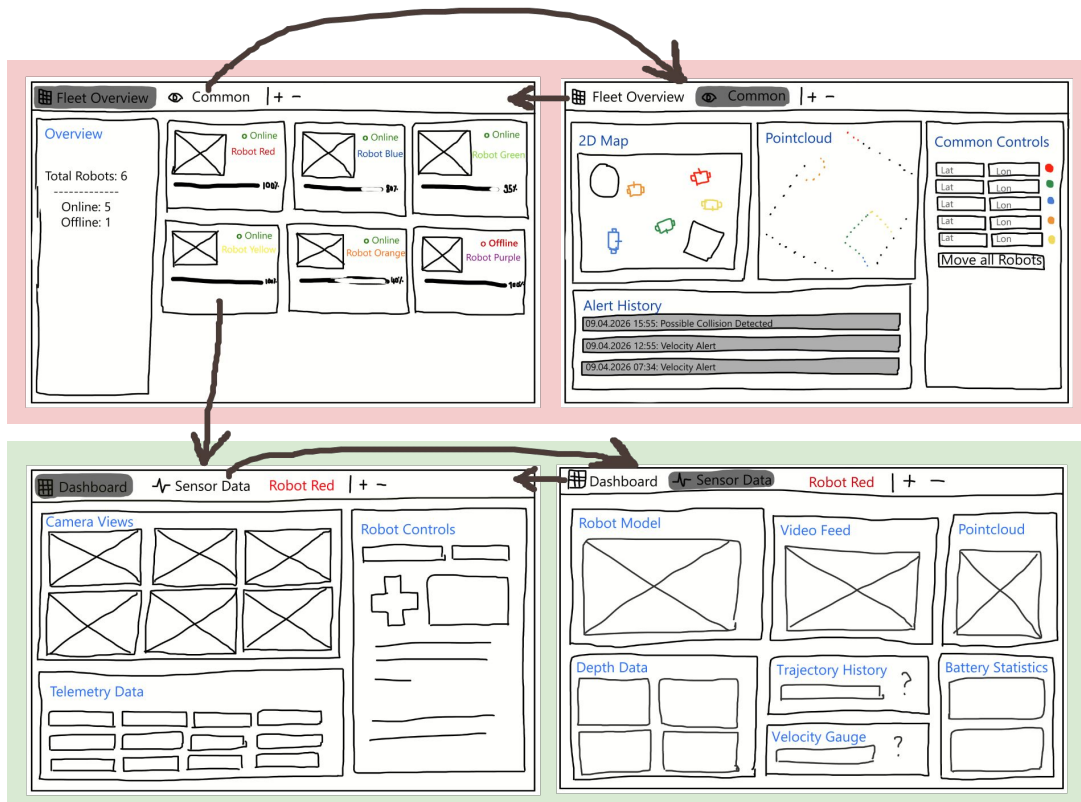
Motivation and Problem

- Autonomous Robots not transparent enough
- **Overview** of all the things going on **inside**
 - Battery, Collision Alerts, Faulty values, Drifting
 - Via some kind of **monitoring software on the web**
- But how does this work when multiple robots are in action?
 - We can't watch all dashboards at once...

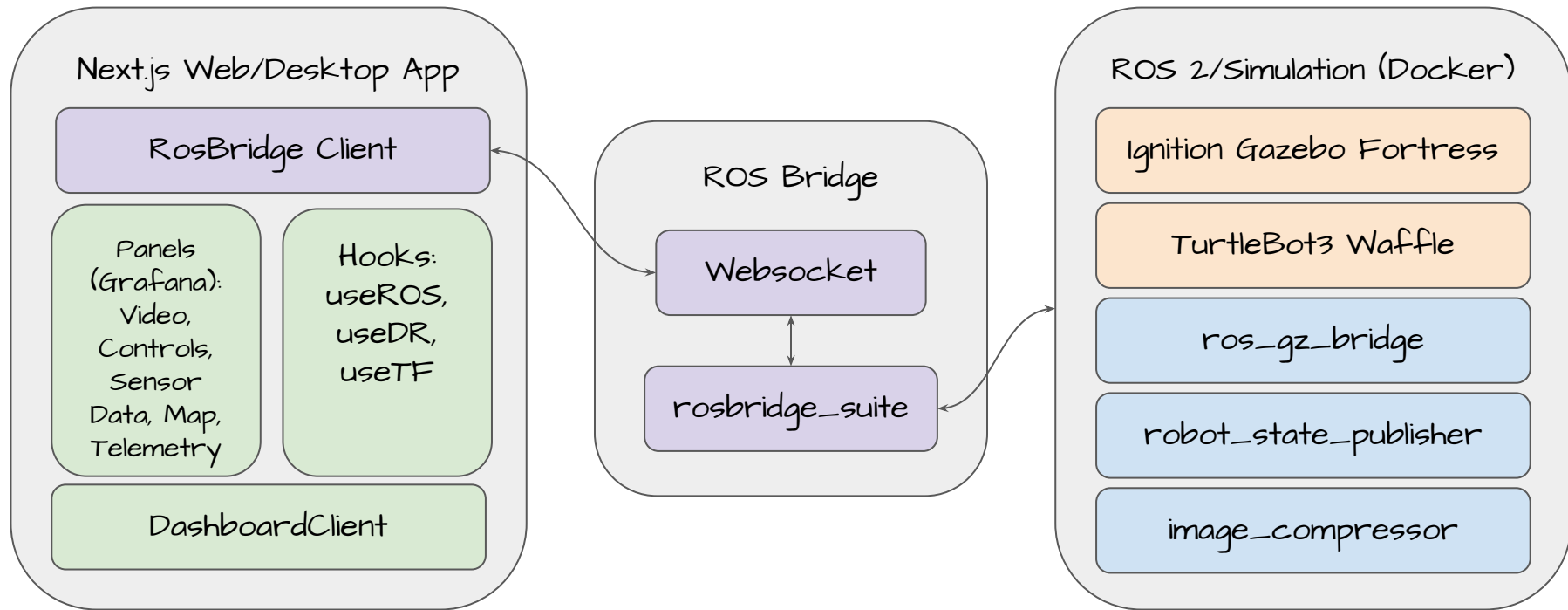


Goal

- Centralized information source, easy to set up
- Separate
 - Common information (Fleet Overview)
 - Robot specific information (Individual Dashboards)
- Combine Robot Info when appropriate



Architecture and Techstack



DEMO

Limitations

- Cross-robot panels on the Common tab (2d map)
- Trajectory history sensor widget
- Camera feed with object detection

Conclusion

- Single centralized interface to watch a fleet telemetry
- Not only monitoring, also robot control
- Modular and easy to extend (add robot or sensors with a minimal setup)
- Work in simulation, need to be tested on real bots

